INSY7315 · TASK 3 · EXPO

SizaniConnect

One record for a food relief network: who is registered, what each household received, which stock it came from, and who was on shift.

Group 13 · Byron Bousfield (ST10445541)

Client: Sizani Community Food Network NPC (case-study client) · Release v1.1.0

sizaniconnect-bb7315.azurewebsites.net

github.com/ByronBousfield1/sizaniconnect

THE PROBLEM

A food network running on paper and WhatsApp

1,400
households on the register

6
distribution sites and a depot

60
volunteers in a month

Two parcels, one household. Paper registers per site cannot catch a household collecting twice.

Hours on WhatsApp. Coordinators confirm every shift by message; no record of hours.

Stock nobody can see. Delivery notes in a file; no one knows what expires next week.

Donor reports take weeks. Figures are reconciled by hand, long after the fact.

THE SOLUTION

One record, captured where the work happens

Site coordinator

Registers households, prints their cards, runs distribution days at their own site

Volunteer

Picks shifts, checks in, scans cards at the event desk on a phone

Operations manager

Verifies households, schedules events, takes in donations, watches stock and expiry

Donor

Sees what their goods reached and prints receipts

Administrator

Manages accounts, offboards leavers, removes old personal data, reads the audit trail

The rules, enforced

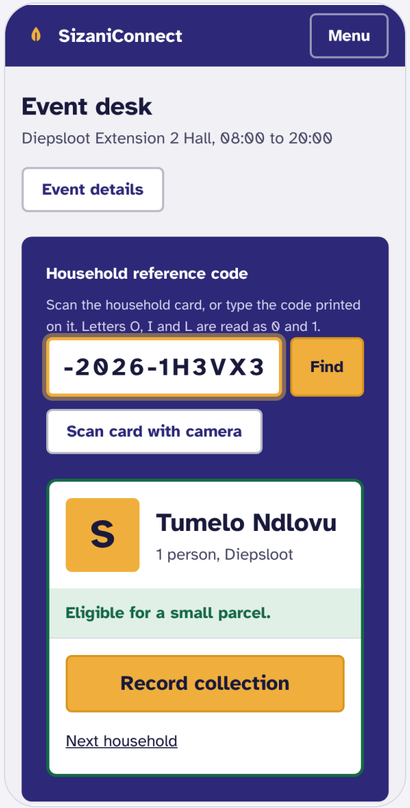
One parcel per household per event, oldest stock first, every change audited

THE EVENT DESK

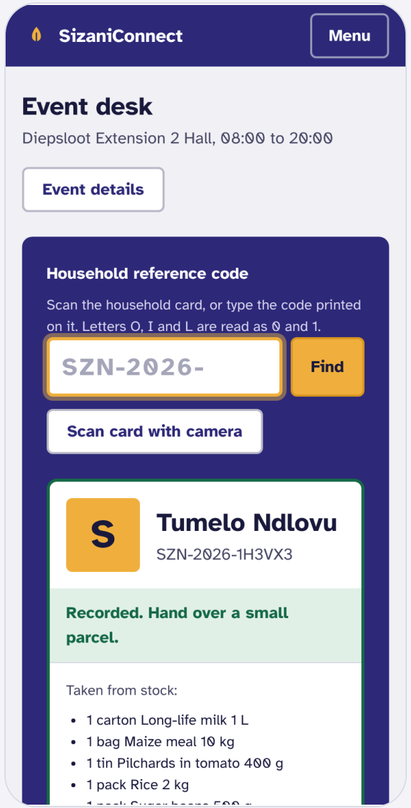
One field, one button, a clear verdict

A volunteer scans the household card or types its code. The desk says what to do in words, not codes.

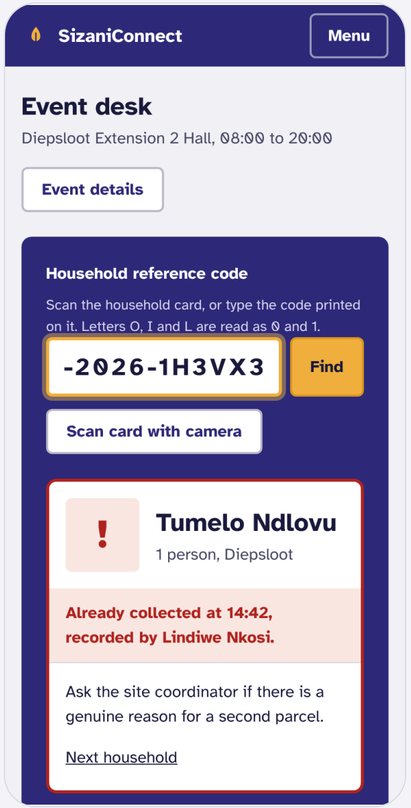
- Eligible: the parcel size to hand over
- Recorded: stock taken oldest expiry first
- Already collected: when, and by whom



Eligible



Recorded



Already collected

TRY IT

Be a volunteer for two minutes

1. Scan a household card: eligible, record, hand over
2. Scan it again: already collected
3. Scan the creased card: explained, nothing sent
4. Flight mode on, record: the banner and a plain message
5. On the laptop: your collection in the report and the audit trail

Open it on your own phone

volunteer@sizani.test
Sizani-Demo-2026

WHAT THE FEEDBACK CHANGED

Eight points of feedback, eight changes, each with a test

The lecturer asked	SizaniConnect now
What if a QR scan is corrupted, on a cheap phone?	Cards carry a check character; bad scans explained, never sent
Show contention and release candidates in the plan	A levelled one-person plan with RC-1 to RC-3
A network timeout in the TLS handshake?	20-second limit, safe repeat, a connection banner
Storage full, database lost: what then?	A plain 503 that says if anything was saved; recovers by itself
Rolling back a destructive migration?	Down scripts, backup tables, a rollback rehearsed in CI
Are cellphone numbers validated?	One format, checked in the browser, the API and the database
The admin forgets the password?	Recovery codes, reset by another admin, audited break-glass
A user retires: removed completely?	Offboard at once, erase after; old households anonymised

FEEDBACK F1 · US-09

A card that survives a bad scan

SIZANI COMMUNITY FOOD NETWORK

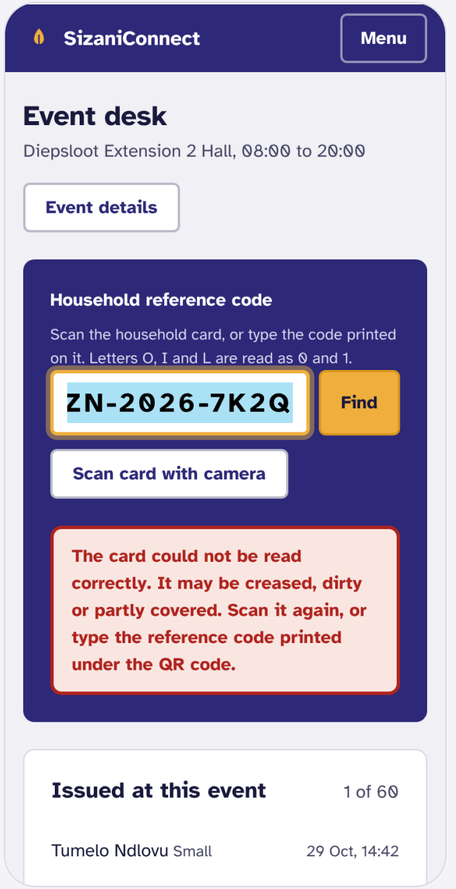
Busisiwe Nkosi

Collects at Tembisa Ivory Park Hall

SZN-2026-415NSP

Bring this card every time. Lost it? Ask your site coordinator for a new one.

The QR code holds SZN1, the code and a check character, so a creased card, someone else’s QR code or a half-typed code is caught on the phone and never sent.



About 2 s to read on a slow phone

FEEDBACK F3 AND F4

When the signal drops, the desk tells the truth

A · HANDSHAKE TIMES OUT

Nothing reached the server

Given up after 20 seconds. A yellow banner; press Record again when the signal is back.

B · SAVED, ANSWER LOST

Pressing again is safe

The database allows one parcel per household per event. The repeat says “your first try was saved”.

C · DATABASE DOWN OR FULL

A plain answer, never a crash

503 with “nothing was saved, try again in a minute”. Recovers by itself, no restart.

Tested by putting the real system into each state: database unreachable, read-only, connections killed mid-save, a request that hangs.

FEEDBACK F6, F7 AND F8

People come, forget passwords and leave

An administrator locked out

1. Email reset link
2. One of ten recovery codes
3. A reset link from another administrator
4. Break-glass from the Azure console, audited

A volunteer who retires

1. Offboard: access stops at once, shifts freed
2. Reversible for 30 days
3. Erase: deleted, or “Former user” if history must stay
4. Households archived 24 months: anonymised

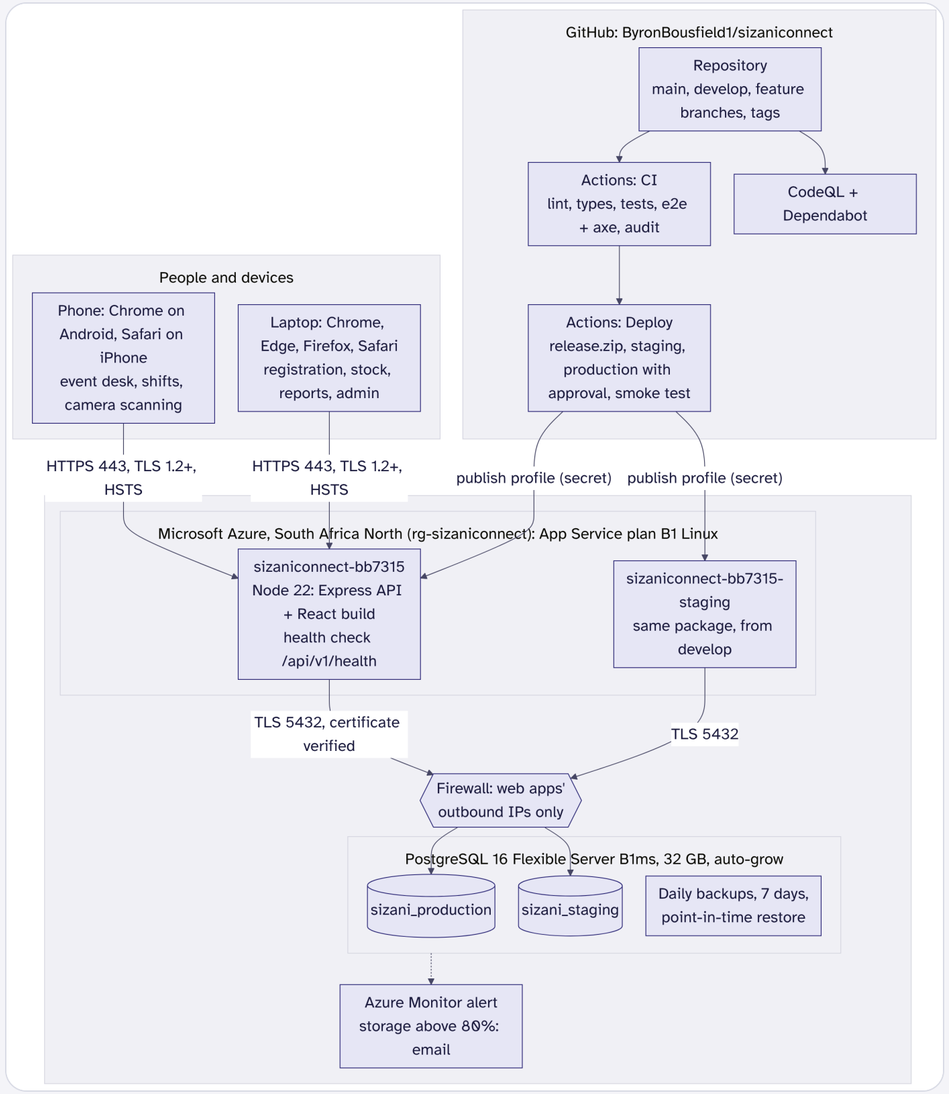
Cellphones 082 555 0134, +27 82 555 0134 and 27825550134 all become +27825550134. Landlines and wrong lengths are refused, by the browser, the API and the database.

HOW IT IS BUILT

A layered web app on Azure, in South Africa

Layer	Built with
Front end	React 19 and TypeScript, one app for phone and laptop
API	Node.js 22, Express 5, Zod validation, JWT sessions
Data	PostgreSQL 16, parameterised SQL, four versioned migrations
Hosting	Azure App Service B1 and PostgreSQL Flexible Server
Delivery	GitHub Actions: CI, CodeQL, staging, approved production

Repository, Unit of Work, Strategy, Observer and state machines keep the rules in one place and testable.



TESTED, MEASURED, DEPLOYED

Nothing reaches production untested

268

automated tests on every push: 209 API, 33 component, 26 browser

96.9%

API code covered by tests (the gate is 70%)

0.43 s

search of 5,000 households at the 95th percentile (target 2 s)

AA

WCAG 2.1 checks on every screen, every role, desktop and phone

Every push: lint, types, tests against real PostgreSQL, browser journeys, accessibility, dependency audit and CodeQL. Staging deploys itself; production waits for an approval and a health check that names the exact build.

Thank you

WHAT COMES NEXT

- A usability session with real volunteers
- Offline capture for sites with no signal
- Delivery-note uploads for donations
- Firefox and Safari in the automated tests

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